Name _____ Date _____
Class _____

Notes: Thermal Energy and Heat

1. Thermal energy is the
   _____
   _____.
2. _____ has thermal energy because
   _____ _____.
3. _____ have the least thermal energy, then _____, and _____ have the most thermal energy.

Stop and Jot. What happens when two objects that have different amounts of thermal energy touch?

Question: How does thermal energy move?

Hypothesis: If a substance with a lot of thermal energy is in contact with a substance with less thermal energy, then what do you think will happen? Why?

Materials: • 1 Ziploc bag • 1 beaker • ice water • hot water • 2 thermometers

Methods:
1. Seal one thermometer tightly into the plastic bag.
2. Place the plastic bag into the beaker of ice water.
3. Every minute, record what you observe in your system and record the temperature on each thermometer. Be sure you record the temperatures in the correct boxes!

Time (min)	**Temperature in bag (°F)**	**Temperature in beaker (°F)**	**Observations** (What does the bag look like? What does the beaker look like? Any changes?)
1			
2			
3			
4			
5			
6			
7			

8			
9			
10			

Conclusion:

1. What happened to the temperature in the bag over time?

2. What happened to the temperature in the beaker?

3. What do you think happened for these changes occur?

4. How do you think the thermal energy moved in this experiment?

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4. When two objects touch, _____ is transferred from one to the other.
5. The amount of energy transferred is called _____.
6. _____ (or _____) always moves from the _____ object to the _____.

Directions: For example example below, draw a picture and arrows to show which way the heat would move.

Hot air balloon lifting off the ground.	Putting your hands under a lamp to warm them up.

Tanning in the sun.	A seat getting warmer when someone sits in it.